

D Weight decay for large language models: additional figures and details

We present the following additional figures related to the LLM experiments. We show that the validation loss of a GPT-2-124M model is determined by the training loss and not influenced by λ in Fig. 19. We also show that the generalization gap stays close to zero throughout training for different λ for both 124M and 774M parameter models. We show the results for models trained *weight decay on LayerNorm weights* in Fig. 20. We see that penalizing all parameters in weight decay (i.e., including the LayerNorm parameters) leads to the same effect for smaller λ (like 0.1) but underperforms on larger λ (like 0.3). Note that when WD is applied on all weights, this changes the optimal value of the objective. In Fig. 21, we train models with ℓ_2 regularization instead of decoupled weight decay as in AdamW (Loshchilov & Hutter, 2019). We observe that ℓ_2 regularization instead of weight decay leads to the same effect as decoupled weight decay (Loshchilov & Hutter, 2019). We train models using SGD with momentum and show the results in Fig. 22. We see that weight decay leads to a similar improvement in training loss for SGD with momentum as well. We show multiple metrics in Fig. 26 for the models shown in Fig. 6: gradient variance, gradient norm, and weight norm plots that complement Fig. 7 in the main part. In Fig. 23, we show results of weight averaging that suggests the suboptimality gap between runs with different λ is much smaller than what the loss at w_t suggests. However, weight averaging is still less effective than fine-tuning with a tiny LR as in Fig. 6. Finally, in Fig. 25, we show results of models trained context length 1024. We see that the training loss over iterations for models trained with a range of LR and WD (all are `bf16`). All runs with LR smaller than 0.001 successfully converge but the final training loss is higher than for LR 0.001. In addition, we observe that lower learning rates prevent the weights from growing too much.

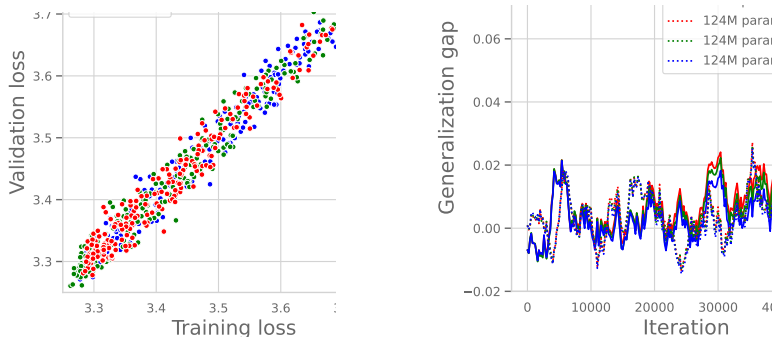


Figure 19: **Left:** The validation loss of a GPT-2-124M model is determined by the training loss and not influenced by λ . **Right:** The generalization gap stays close to zero throughout training for different λ for both 124M and 774M parameter models.

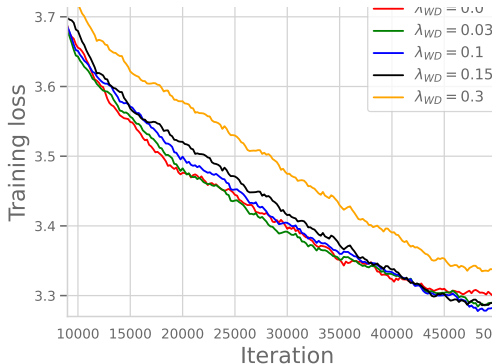


Figure 20: **GPT-2-124M on OpenWebText with weight decay on LayerNorm weights.** Penalizing all parameters in weight decay (i.e., including the LayerNorm parameters) leads to the same effect for smaller λ (like 0.1) but underperforms on larger λ (like 0.3). Note that when WD is applied on all weights, this changes the optimal value of the objective.

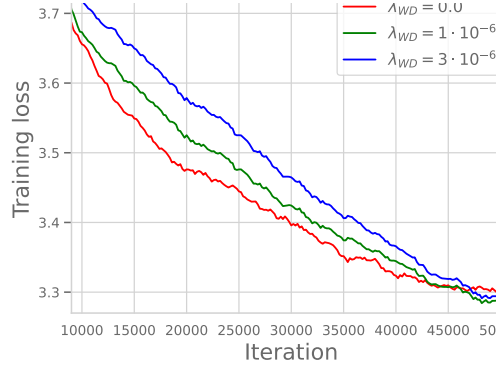


Figure 21: **GPT-2-124M on OpenWebText with ℓ_2 regularization.** We observe that ℓ_2 regularization instead of weight decay leads to the same effect as decoupled weight decay (Loshchilov & Hutter, 2019).

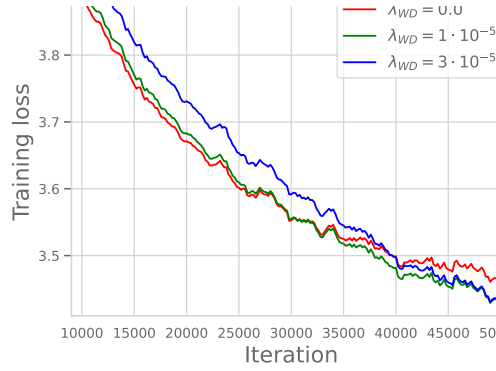


Figure 22: **GPT-2-124M on OpenWebText trained with SGD with momentum.** Weight decay leads to a similar improvement in training loss for *SGD with momentum* as well (all other experiments are done with AdamW).

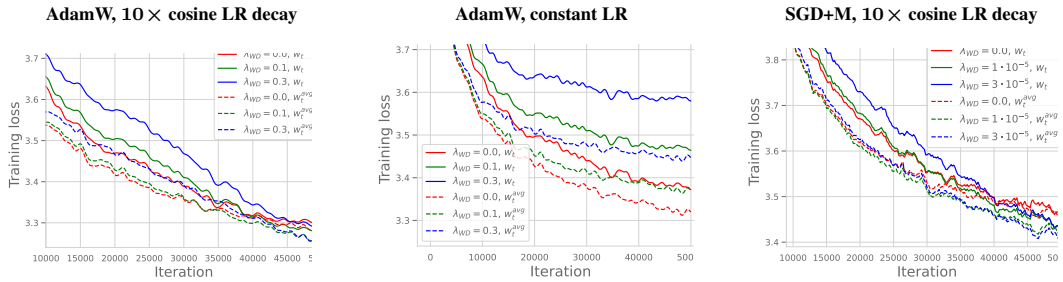


Figure 23: **Weight averaging for GPT-2-124M on OpenWebText.** Weight averaging (w_t^{avg}) shows that the suboptimality gap between runs with different λ is much smaller than what the loss at w_t suggests. However, weight averaging is still less effective than fine-tuning with a tiny LR as in Fig. 6.

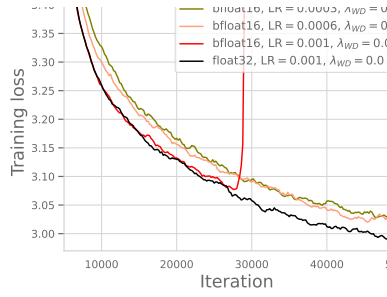


Figure 24: **GPT-2-124M on OpenWebText with context length 1024.** The model trained with a moderate LR 0.001 diverges for bfloat16 but not for float32; lowering the LR prevents the divergence but leads to a worse loss.

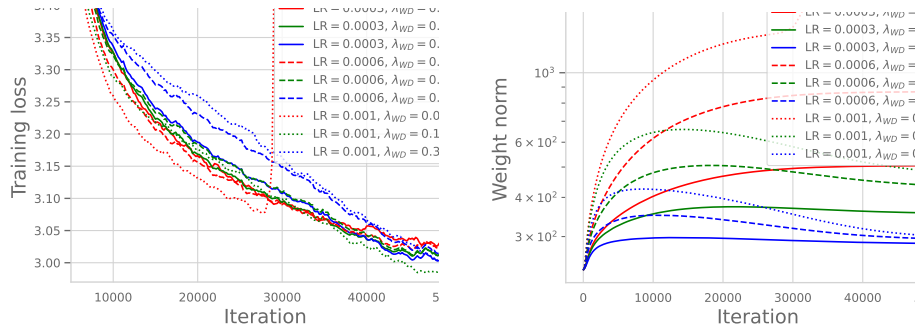


Figure 25: **GPT-2-124M on OpenWebText with context length 1024.** (Left) The training loss over iterations for models trained with a range of LR and WD (all are bfloat16). All runs with LR smaller than 0.001 successfully converge but the final training loss is higher than for LR 0.001. (Right) Weight norms for LR in 0.0003, 0.0006, 0.001 for $\lambda = 0.1$ which does not diverge. Lower learning rates prevent the weights from growing too much.

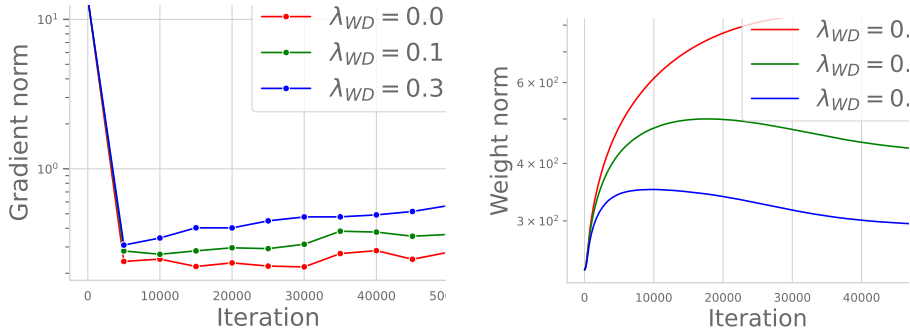


Figure 26: **GPT-2-124M on OpenWebText.** The gradient norm and weight norm plots for the models reported in Fig. 7.